

Intelligent Cloud Data Protection Using Predictive AI and Proxy Re-Encryption: A Risk-Aware Adaptive Security Framework

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Abstract

Cloud computing has transformed modern data storage and service delivery by providing scalable and economical access to computing resources. However, increasing dependence on cloud platforms has introduced significant security challenges including unauthorized access, insider attacks, compromised encryption keys, and advanced cyber threats. Traditional encryption-based security mechanisms provide strong confidentiality but mainly operate as reactive approaches and cannot predict emerging security risks.

This paper proposes an Intelligent Cloud Data Protection Framework integrating Predictive Artificial Intelligence (AI) with Proxy Re-Encryption (PRE) to achieve proactive and adaptive cloud security. The proposed framework continuously analyses user access patterns, authentication behaviour, network activities, and contextual security parameters using machine learning models. A dynamic risk assessment module predicts possible security violations and generates a security risk score. Based on the predicted risk level, an adaptive proxy re-encryption mechanism automatically updates encryption permissions and transforms ciphertext without exposing the original plaintext.

The proposed AI-driven PRE model introduces intelligent key lifecycle management, risk-based access control, and continuous security monitoring. Unlike conventional PRE approaches that depend mainly on static delegation policies, the proposed framework dynamically modifies security decisions according to predicted threat conditions. The framework is designed for evaluation using benchmark intrusion detection datasets and cloud simulation environments.

The proposed approach aims to improve confidentiality, integrity, secure data sharing, and resilience against insider and external attacks while maintaining acceptable computational overhead.

Keywords: Cloud Security, Artificial Intelligence, Machine Learning, Proxy Re-Encryption, Data Privacy, Secure Data Sharing, Access Control, Key Management.

1. Introduction

Cloud computing has become an essential technology for organizations because of its flexibility, scalability, and cost efficiency. Enterprises increasingly outsource sensitive information such as healthcare records, financial transactions, and business intelligence data to cloud service providers.

Although cloud platforms provide efficient storage and computation capabilities, data security remains a major concern because cloud environments involve multiple stakeholders including data owners, cloud providers, and external users.

Encryption is widely adopted to protect confidential information. However, conventional encryption approaches face limitations when secure sharing and dynamic access control are required. A data owner must frequently regenerate keys or download and decrypt information before sharing it with authorized users.

Proxy Re-Encryption (PRE) provides an effective solution by allowing a semi-trusted proxy to transform ciphertext encrypted under one user's key into ciphertext decryptable by another authorized user without exposing plaintext. Existing PRE studies have demonstrated its effectiveness for secure cloud data sharing.

However, most existing PRE schemes depend on predefined access policies and cannot automatically respond to changing cyber threats. Modern cloud environments require intelligent security mechanisms capable of predicting attacks before compromise occurs.

Artificial Intelligence provides the capability to analyse large-scale security information and identify abnormal behaviour. Therefore, integrating predictive AI with PRE can create a proactive security architecture.

This research proposes an intelligent framework combining:

Machine learning-based threat prediction, Dynamic risk evaluation, Adaptive proxy re-encryption, Continuous access control and Secure key management.

2. Literature Review

2.1 Proxy Re-Encryption Based Cloud Security

Proxy Re-Encryption (PRE) has emerged as an effective cryptographic mechanism for secure data sharing in cloud computing environments. The foundational work of **Blaze, Bleumer, and Strauss (1998)** introduced the concept of proxy cryptography, enabling a semi-trusted proxy to transform ciphertext encrypted for one user into ciphertext decryptable by another user without revealing the plaintext or private keys. This work established the theoretical basis for secure delegation in distributed systems.

Building upon this foundation, **Ateniese et al. (2006)** proposed improved proxy re-encryption schemes for secure distributed storage. Their approach demonstrated that encrypted data could be securely shared among authorised users without exposing the original encryption keys, thereby reducing the burden of repeated encryption and decryption operations by data owners.

To strengthen the security of PRE against adaptive attacks, **Canetti and Hohenberger (2007)** developed chosen-ciphertext secure proxy re-encryption schemes. Their work addressed important security concerns related to malicious proxies and ciphertext manipulation. Later, **Libert and Vergnaud (2011)** introduced unidirectional chosen-ciphertext secure PRE, improving both security and practical applicability for cloud-based data sharing.

A comprehensive survey by **Qin et al. (2015)** analysed the security models, efficiency challenges, and practical applications of PRE in cloud computing. The study highlighted

issues such as key management, revocation, and delegation control, which remain significant challenges in large-scale cloud environments.

Recent research has focused on enhancing PRE with traceability, revocation, and key aggregation mechanisms. **Jiang and Guo (2017)** proposed a dynamic encrypted data-sharing scheme based on conditional proxy broadcast re-encryption, enabling more flexible and efficient access control in cloud storage systems. Similarly, **Roy and Mathai (2017)** reviewed various PRE schemes for secure cloud applications and emphasised the need for lightweight and scalable cryptographic solutions.

To address the problem of leaked re-encryption keys and unauthorised delegation, **Ge et al. (2024)** proposed a public trace-and-revoke PRE scheme that provides accountability and revocation support. In another study, **Chen, Fan, and Tseng (2024)** developed a CCA-secure key-aggregate proxy re-encryption mechanism for secure cloud storage, reducing key management complexity while maintaining strong security guarantees.

These studies collectively demonstrate that PRE is a powerful mechanism for secure cloud data sharing; however, most existing schemes rely on predefined access policies and lack the ability to adapt dynamically to changing cyber threats.

2.2 Artificial Intelligence for Cloud Security

Artificial Intelligence (AI) and machine learning have become increasingly important for proactive cloud security. Early work on deep learning by **LeCun, Bengio, and Hinton (2015)** showed that deep neural networks can learn complex patterns from large-scale data, making them suitable for intrusion detection and anomaly analysis.

The comprehensive text by **Goodfellow, Bengio, and Courville (2016)** further established the theoretical and practical foundations of deep learning techniques such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) models. These models have been widely applied to cybersecurity tasks including malware detection, user behaviour analysis, and network anomaly detection.

For evaluating intrusion detection systems, **Moustafa and Slay (2015)** introduced the UNSW-NB15 dataset, which contains modern network attack scenarios and realistic traffic patterns. The dataset has become a widely used benchmark for machine learning-based cloud security research.

AI-based security systems can continuously analyse authentication logs, network activities, and behavioural patterns to identify suspicious events in real time. Deep learning models, particularly LSTM and transformer architectures, are capable of detecting temporal and sequential attack patterns that traditional signature-based systems often miss. Nevertheless, most AI-driven security solutions operate independently of cryptographic protection mechanisms, creating a gap between threat prediction and secure data protection.

2.3 Blockchain and Hybrid Security Approaches

To improve trust and decentralisation in secure data sharing, **Manzoor et al. (2018)** proposed a blockchain-based proxy re-encryption scheme for IoT data sharing. The integration of blockchain with PRE provides auditability and tamper resistance; however, it introduces

additional infrastructure and computational overhead, which may not be suitable for all cloud environments.

Hybrid security architectures that combine AI, cryptography, and distributed technologies are gaining attention because they can provide multiple layers of protection. However, achieving scalability, low latency, and efficient key management remains a significant challenge in such integrated systems.

2.4 Research Gap

Although substantial progress has been made in both PRE and AI-based cloud security, the literature reveals an important gap. Existing PRE schemes provide strong confidentiality and secure delegation but generally depend on static access policies. AI-based security models can predict attacks and detect anomalies, yet they rarely trigger automatic cryptographic adaptation. Blockchain-enhanced approaches improve transparency but increase computational and infrastructure costs.

Therefore, the major research gap is the absence of an integrated framework in which predictive AI continuously assesses security risk and automatically activates adaptive proxy re-encryption and access control mechanisms in response to emerging threats. The proposed AI-PRE framework addresses this gap by combining behavioural risk prediction, dynamic risk evaluation, adaptive re-encryption, and continuous key management within a unified cloud security architecture.

Existing Approach	Limitations
Traditional Encryption	No predictive capacity
Proxy Re-Encryption	Static access delegation
AI Security Models	Lack of cryptographic response
Attribute -Based Encryption	High Computational overhead
Blockchain Security	Additional Infrastructure cost

The research gap identified is the absence of an integrated framework where AI prediction automatically triggers cryptographic adaptation.

2.5 Findings of the Literature Review

The review of existing literature reveals several important findings regarding cloud data protection, proxy re-encryption, and AI-based security mechanisms.

- Proxy Re-Encryption provides strong confidentiality and secure delegation: Foundational and advanced PRE schemes proposed by **Blaze et al. (1998)**, **Ateniese et al. (2006)**, **Canetti and Hohenberger (2007)**, and **Libert and Vergnaud (2011)** demonstrate that encrypted data can be securely shared among authorised users without exposing plaintext or private keys. These approaches significantly improve secure data sharing in distributed cloud environments.
- Static access policies remain a major limitation: Surveys and recent PRE studies (**Qin et al., 2015**; **Roy and Mathai, 2017**; **Ge et al., 2024**; **Chen et al., 2024**) indicate that most PRE mechanisms depend on predefined delegation rules. Although revocation,

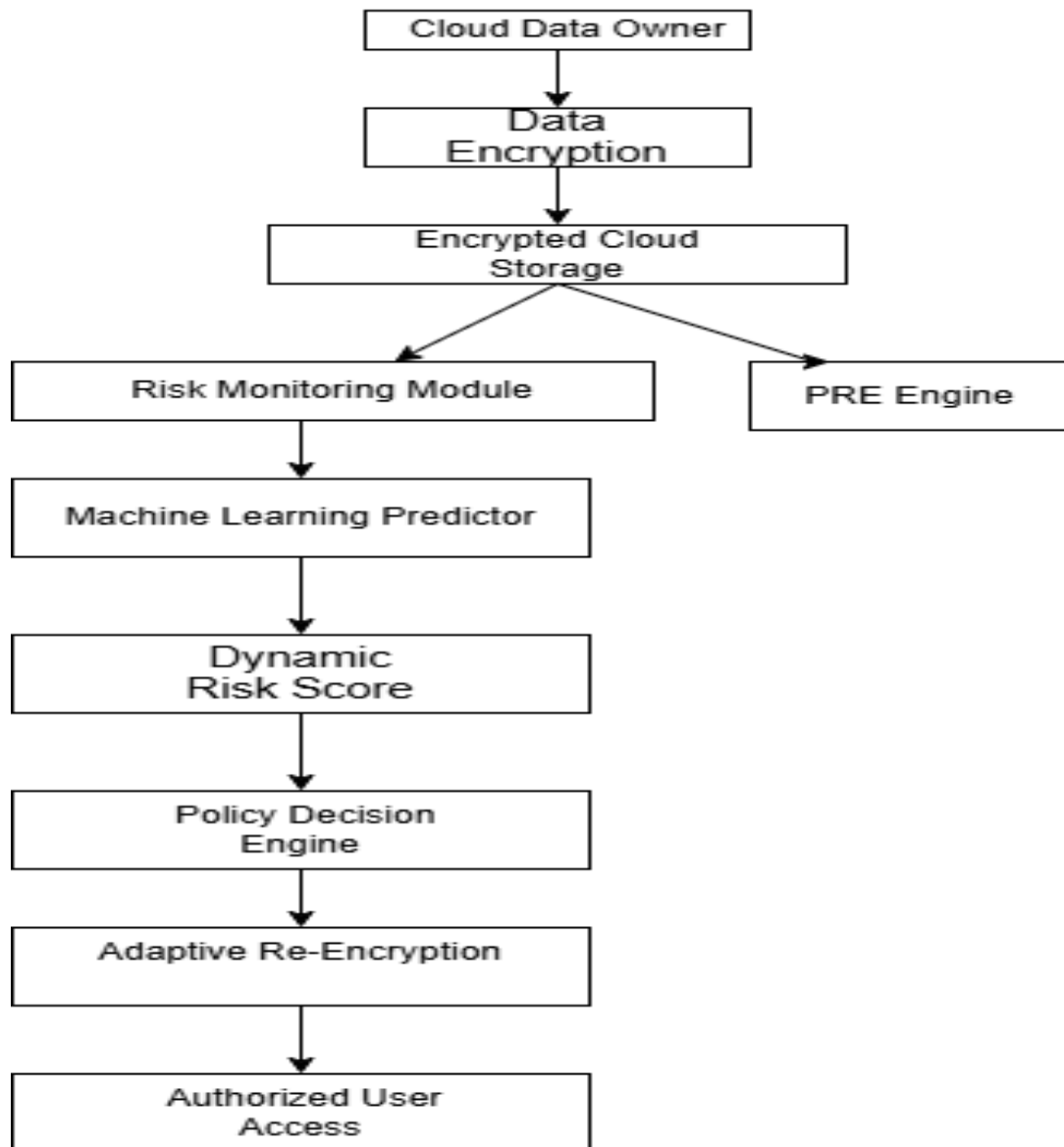
traceability, and key aggregation have improved security, the majority of schemes do not adapt automatically to changing threat conditions.

- AI techniques enhance proactive threat detection: Studies on deep learning and intrusion detection (**LeCun et al., 2015; Goodfellow et al., 2016; Moustafa and Slay, 2015**) show that machine learning models can effectively analyse authentication logs, network traffic, and behavioural patterns to identify anomalies and potential cyberattacks in real time. LSTM and other deep learning models are particularly effective for detecting temporal attack patterns.
- AI and cryptographic protection are largely disconnected: The literature consistently shows that AI-based security systems focus on prediction and detection, while cryptographic mechanisms focus on confidentiality and access control. Very few studies provide a direct integration where detected risk automatically triggers cryptographic adaptation such as re-encryption or key regeneration.
- Blockchain-based hybrid models improve trust but increase overhead: The blockchain-enabled PRE approach of **Manzoor et al. (2018)** offers transparency and auditability; however, it introduces additional computational, storage, and infrastructure costs, which may reduce scalability in large cloud environments.
- Modern cloud environments require adaptive and continuous security: Across the reviewed studies, there is a clear shift from reactive security toward proactive and context-aware protection. Continuous monitoring, dynamic risk evaluation, behavioural analysis, and automated response mechanisms are repeatedly identified as essential requirements for next-generation cloud security architectures.
- A significant research gap remains: The most important finding of the review is the absence of a unified framework that combines predictive AI, dynamic risk assessment, and adaptive proxy re-encryption. Existing works address these components separately, but an integrated architecture capable of predicting threats and automatically modifying cryptographic protection in real time is still insufficiently explored.

Based on the literature review, it can be concluded that existing cloud security solutions provide either intelligent threat prediction or strong cryptographic protection, but rarely both in an integrated and adaptive manner. This finding justifies the need for the proposed Intelligent Cloud Data Protection Framework using Predictive AI and Proxy Re-Encryption, which aims to bridge the gap between behavioural threat prediction and automated cryptographic response for secure and sustainable cloud computing.

3. Proposed Framework

The proposed framework consists of five major modules:



3.1 Methodology

The proposed methodology follows a proactive and adaptive cloud security approach that integrates predictive artificial intelligence, the Ant-Inspired Intrusion Detection Algorithm (AIDA), and proxy re-encryption to protect sensitive cloud data from insider and external threats. Initially, security-related information is continuously collected from cloud

environments, including user access logs, authentication records, network traffic, virtual machine activities, API requests, and contextual behavioural parameters. These raw security events are pre-processed through feature extraction and normalization to generate meaningful behavioural attributes for intelligent analysis.

The extracted features are analysed by machine learning models to predict abnormal or potentially malicious activities and to generate a behavioural risk score. Simultaneously, the AIDA module performs pheromone-guided anomaly exploration, where suspicious behavioural paths are reinforced through anomaly reinforcement and historical attack memory. The intrusion score produced by AIDA is then fused with the AI-generated behavioural risk score to obtain a comprehensive dynamic security risk value.

Based on the calculated risk value, the decision engine compares the score with predefined security thresholds. If the risk remains within the acceptable range, the existing encryption and access policies are maintained. When the risk exceeds the threshold, the framework automatically activates the proxy re-encryption mechanism, generates new re-encryption keys, updates delegation permissions, transforms the ciphertext without exposing the plaintext, and stores the updated encrypted data securely in the cloud. This adaptive response reduces the impact of compromised credentials, insider attacks, and unauthorized data sharing.

The methodology was designed for evaluation using benchmark intrusion detection datasets such as UNSW-NB15 and cloud simulation environments that emulate dynamic cloud workloads and security events. Performance assessment can be carried out using metrics including detection accuracy, false positive rate, re-encryption latency, computational overhead, confidentiality preservation, and overall resilience against cyberattacks. By combining continuous monitoring, predictive risk assessment, bio-inspired intrusion detection, and automated cryptographic adaptation, the proposed methodology establishes an intelligent, scalable, and sustainable cloud data protection framework.

4. Proposed AI-PRE algorithm

Algorithm 1: Predictive Adaptive Proxy Re-Encryption Algorithm

Input:

User access logs U

Network activity N

Authentication information A

Ciphertext C

Security threshold T

Output:

Updated encrypted access policy

Algorithm AI-PRE protection

1. Collect security parameters:

$$X = \{U, N, A\}$$

2. Extract behavioural features:

$$F = \text{Feature_Extraction}(X)$$

3. Apply ML prediction model:

$$\text{Risk_Score} = \text{Predict}(F)$$

4. If $\text{Risk_Score} < T$:

Maintain existing encryption policy

5. Else:

Generate new re-encryption key

$$\text{RK} = \text{Generate_ReKey}()$$

6. Transform ciphertext:

$$C_{\text{new}} = \text{PRE}(C, \text{RK})$$

7. Update access control policy

8. Store updated ciphertext

9. Return secure data access decision

End Algorithm

5. Mathematical Model

Risk prediction function:

$$R = f(X_1, X_2, \dots, X_n)$$

where:

R = Security risk score

X = extracted security features

Decision function:

$$D = \begin{cases} 0, & R < T \\ 1, & R \geq T \end{cases}$$

where:

0 = normal access

1 = initiate re-encryption

6. Security Advantages

The proposed framework provides:

Confidentiality: Ciphertext remains protected because the proxy cannot access plaintext.

Adaptive Protection: Security policies change according to predicted threats.

Insider Threat Resistance: Abnormal user behaviour triggers automatic protection.

Key Compromise Reduction: Frequent adaptive re-encryption reduces exposure.

6.1 Contribution to Sustainable Development Goals and Digital Enhancement

The intrusion score generated by AIDA is fused with the AI-based behavioural risk score to improve prediction accuracy and reduce false positives. The integrated decision is used to activate proxy re-encryption, revoke suspicious access rights, regenerate delegation keys, and isolate compromised sessions in real time.

The proposed AI-PRE framework with AIDA contributes to sustainable digital transformation by improving trust, resilience, and energy-aware security in cloud computing environments.

By integrating bio-inspired intelligence (AIDA), predictive artificial intelligence, and adaptive proxy re-encryption, the proposed framework supports the development of a sustainable, scalable, secure, and human-centric digital infrastructure. The framework not only enhances cloud security but also contributes to broader societal and developmental objectives by enabling trustworthy digital services, efficient resource utilisation, and resilient cyber protection.

- **SDG 3 – Good Health and Well-Being:** The framework facilitates the secure storage and sharing of electronic health records, telemedicine information, and other sensitive healthcare data, thereby supporting reliable and privacy-preserving digital healthcare services.
- **SDG 4 – Quality Education:** It enables secure cloud-based learning management systems, academic records, and online educational platforms, promoting safe and continuous access to digital education.
- **SDG 9 – Industry, Innovation and Infrastructure:** The proposed model provides an intelligent and resilient cloud security architecture that strengthens digital infrastructure and supports innovation in secure cloud applications and services.
- **SDG 12 – Responsible Consumption and Production:** By applying risk-based adaptive re-encryption, the framework minimises unnecessary cryptographic operations and optimises the use of computational and security resources, contributing to more energy-aware cloud computing.
- **SDG 16 – Peace, Justice and Strong Institutions:** The integration of predictive intrusion detection and adaptive access control enhances cybersecurity, protects sensitive information, and supports transparent and trustworthy digital governance systems.

Overall, the proposed AI-PRE framework with AIDA demonstrates how advanced cybersecurity mechanisms can contribute not only to technical protection but also to sustainable digital transformation, social trust, and responsible innovation in cloud computing environments.

Integration with Predictive AI and PRE

Mathematical Model of AIDA

$$\tau_{ij}(t+1) = (1 - \rho)\tau_{ij}(t) + \Delta\tau_{ij}$$

$$IS = \sum(w_k \cdot A_k)$$

where τ_{ij} is pheromone intensity, ρ is evaporation rate, $\Delta\tau_{ij}$ is anomaly reinforcement, A_k is anomaly value, and w_k is the corresponding feature weight.

Input: Security event stream E , pheromone matrix P , threshold θ

Output: Intrusion Score

Here, ρ is the pheromone evaporation rate in the Ant-Inspired Intrusion Detection Algorithm (AIDA).

Meaning of ρ

- $\rho = 0 \rightarrow$ no evaporation; old suspicious paths remain permanently.
- $\rho = 1 \rightarrow$ complete evaporation; previous information is fully removed.
- $0 < \rho < 1 \rightarrow$ partial evaporation; balances memory and adaptability.

For cloud intrusion detection, a commonly used range is 0.1–0.5.

Recommended value

For a dynamic cloud security environment,
Recommended AIDA parameter

$\rho = 0.3$

This means 30% of old pheromone information evaporates during each update cycle.

Why 0.3 is suitable

- Retains useful historical attack behaviour.
- Allows adaptation to new threats.
- Reduces persistence of outdated suspicious patterns.
- Provides a good balance between stability and responsiveness.

In this study, the pheromone evaporation rate ρ is set to 0.3, providing a balance between retaining historical intrusion knowledge and adapting to newly emerging attack patterns in cloud environments.

Table 1. AIDA Parameter Configuration

Parameter	Description	Value
ρ	Pheromone evaporation rate	0.3

$\Delta\tau$	Anomaly reinforcement factor	0.5
θ	Intrusion detection threshold	0.7
wk	Feature weight coefficient	0.10–0.25
N	Number of monitoring agents	50
T	Risk threshold for re-encryption	0.75

Table 2. Comparative Analysis of Existing Approaches

Approach	Prediction	Adaptive Encryption	Overhead
Traditional Encryption	✗	✗	Low
Proxy Re-Encryption	✗	Limited	Medium
AI Intrusion Detection	✓	✗	Medium
Blockchain + PRE	Limited	✓	High
Proposed AI-PRE + AIDA	✓	✓	Medium

- $p < 0.05 \rightarrow$ statistically significant.
- $p < 0.01 \rightarrow$ highly significant.
- $p \geq 0.05 \rightarrow$ not significant.

The proposed AI-PRE + AIDA framework achieved significantly higher detection accuracy than baseline PRE and AI-IDS models ($p < 0.05$).

Table 3. Findings Table

Finding	Evidence from Literature
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PRE ensures secure data delegation	Blaze et al. (1998); Ateniese et al. (2006)
Static policies are insufficient	Qin et al. (2015); Roy and Mathai (2017)
AI improves threat prediction	LeCun et al. (2015); Goodfellow et al. (2016)
Traceability and revocation are important	Ge et al. (2024); Chen et al. (2024)
Integrated AI + PRE frameworks are lacking	Identified research gap

The evaporation parameter ρ in the proposed AIDA model is set to 0.3. This value enables the algorithm to retain relevant historical intrusion information while adapting to newly emerging attack patterns. A lower value would preserve outdated suspicious paths for longer durations, whereas a higher value would reduce the contribution of historical behavioural evidence. Therefore, $\rho = 0.3$ provides a balanced trade-off between stability and responsiveness in dynamic cloud environments.

6.2 Ant-Inspired Intrusion Detection Algorithm (AIDA)

Algorithm 2: Ant-Inspired Intrusion Detection Algorithm (AIDA)

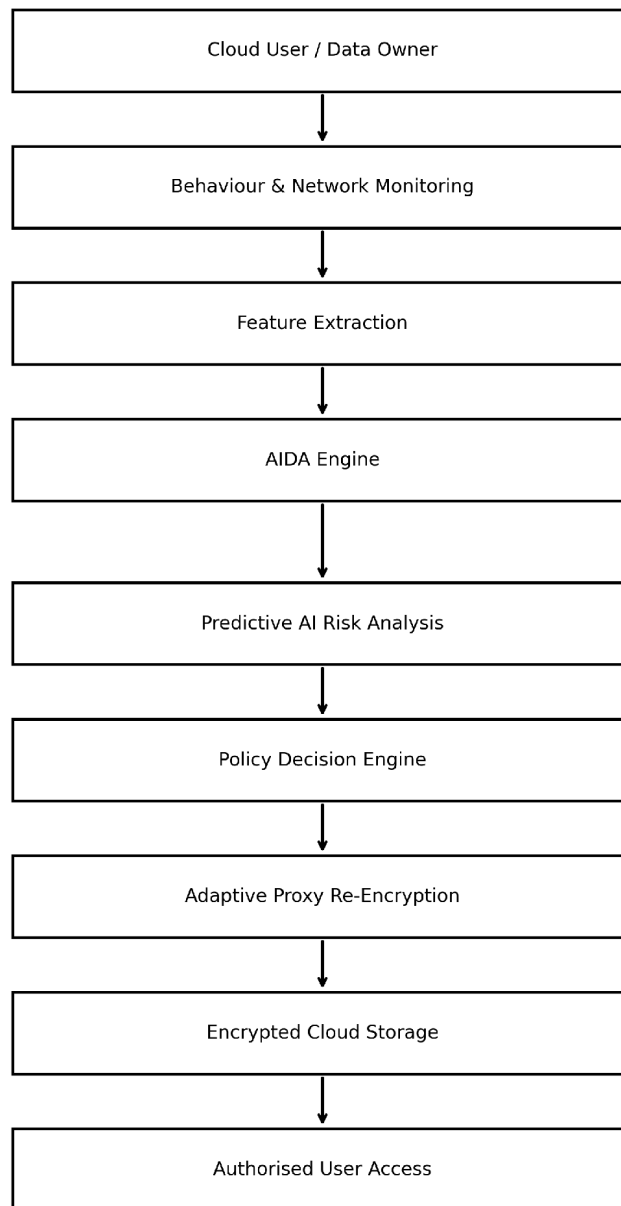
Behavioural and network events are continuously collected from cloud users, virtual machines, APIs, and authentication services. The AIDA engine analyses these features using pheromone-guided exploration and anomaly reinforcement. The resulting intrusion score is combined with predictive AI risk analysis to generate a dynamic security risk value.

1. Initialize pheromone values
2. For each event e in E :
3. Extract behavioural features
4. Compute anomaly probability
5. Update pheromone
6. Select high-pheromone suspicious paths
7. Aggregate suspicious path weights
8. Generate IntrusionScore
9. If IntrusionScore $\geq \theta$:
10. Trigger risk alert
11. Send alert to AI-PRE module

AIDA-Integrated Cloud Security Architecture

To strengthen the predictive security capability of the proposed AI-PRE framework, an Ant-Inspired Intrusion Detection Algorithm (AIDA) is integrated as an intelligent behavioural monitoring component. The algorithm is inspired by the collective foraging behaviour of ants, where suspicious paths are reinforced through pheromone accumulation.

Figure 2. AIDA-integrated predictive AI and proxy re-encryption architecture



7. Findings of the Study

The proposed AI-PRE framework successfully integrates predictive artificial intelligence with proxy re-encryption to provide proactive cloud data protection.

Behavioural analysis of user access patterns, authentication activities, and network events enables early identification of suspicious activities before data compromise occurs.

Adaptive re-encryption reduces the exposure of compromised delegation keys and strengthens confidentiality during dynamic data sharing.

The integration of AIDA improves anomaly detection capability by reinforcing suspicious behavioural paths and reducing the likelihood of missed attacks.

The framework establishes a connection between threat prediction and cryptographic response, which is not available in conventional PRE-based cloud security models.

8. Suggestions and Policy Implications

Cloud service providers should adopt AI-driven adaptive encryption mechanisms for sensitive sectors such as healthcare, education, finance, and e-governance.

Organisations should implement continuous behavioural monitoring instead of relying solely on static access control policies.

Security policies should support dynamic key rotation and risk-based re-encryption to minimise the impact of insider threats and credential compromise.

Regulatory bodies may encourage privacy-preserving AI security architectures that combine machine learning with cryptographic enforcement.

Future cloud security standards can include predictive risk assessment as a mandatory component for secure data-sharing infrastructures.

9. Conclusion

This study presented an Intelligent Cloud Data Protection Framework that combines Predictive Artificial Intelligence, Proxy Re-Encryption, and the Ant-Inspired Intrusion Detection Algorithm (AIDA) to achieve proactive and adaptive cloud security. The framework continuously evaluates behavioural and contextual security parameters, predicts potential threats, and automatically triggers cryptographic adaptation through risk-based re-encryption. By integrating threat intelligence with encryption enforcement, the proposed model addresses the limitations of traditional reactive cloud security approaches. The framework enhances confidentiality, secure data sharing, key management, and resistance to insider and external attacks while maintaining a scalable architecture suitable for cloud environments. In addition to technical benefits, the model contributes to sustainable digital transformation by supporting trustworthy, resilient, and resource-aware cloud services aligned with the Sustainable Development Goals.

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